E-Factor IoT Module – Technical Specification

1. Overview

The E-Factor IoT Module is a compact and versatile connectivity solution designed for intelligent control and data communication across a wide range of IoT applications. It supports multiple communication protocols — Wi-Fi, RF, and Power Line Communication (PLC) — enabling seamless integration with end devices through UART. This module communicates with the E-Factor Cloud and E-Factor Mobile App, ensuring secure, reliable, and real-time data exchange. It supports OTA (Over-the-Air) firmware updates and is optimized for efficient and stable performance in both consumer and industrial IoT environments.

2. Key Features

- Multi-Protocol Communication: Wi-Fi, RF, and PLC support.
- Flexible UART Communication: Baud rates – 4800, 9600, 38400, 57600, 115200 (default), and 230400.
- Cloud & Mobile Connectivity: Compatible with E-Factor Cloud and the E-Factor IoE Mobile App.
- OTA Firmware Update: Seamless over-the-air updates from the E-Factor Cloud.
- Low-Power Operation: Optimized for 3.3 V ± 10% with efficient current draw.
- Compact Design: Suitable for integration into diverse end devices.

3. Electrical Specifications

Parameter	Value	Notes
Supply Voltage	3.3 V DC	± 10% tolerance
Current Consumption	0.5 A (Min)	Varies by protocol
Logic Level	3.3 V	UART compatible

4. UART Communication

Default Baud Rate: 115200 bps

Supported Baud Rates: 4800 – 230400 bps

4.1. Simple Frame (for payload <= 64 bytes)

When the total data length is **64 bytes or less**, the module communicates using a **simple frame format**.

This mode does **not follow a fixed frame structure**; users may define their own data format according to application needs.

- All data is transmitted and received in **hexadecimal byte format** over UART.
- Users can freely define their own sequence (e.g., command, data, checksum, etc.).
- Maximum data per frame: **64 bytes**.
- For payloads exceeding 64 bytes, refer to the **Multi-Frame Format** (Section 4.2).

Example:

AA 05 01 23 45 67 89 BB – user-defined frame structure.

4.2. Multi-Frame (for payload > 64 bytes)

Field	Description	Valid / Example Values	Size
CMD	Internal command identifier for this stream. For data exchange it is constant.	0x2B (Reserved)	1 byte
No. of Frames	Total number of frames in this message when payload is greater than 64 bytes.	01 to FF (example: 0x04 when split into 4 frames)	1 byte
Frame Index	Sequence number of the current frame.	0x00 to N-1 (example: 0x00 for first frame)	1 byte
Data	Payload carried by the frame.	Less than or equal to 64 bytes per frame	0 to 64 bytes

Example

CMD	No. of Frames	Frame Index	Data (<= 64 bytes)
2B	04	00	Data

Note: Maintain a minimum 300 ms gap between consecutive UART frames for reliable transmission and reception.

5. Programming and Boot Interface (SW1 / SW2)

The module provides an easy flashing and debugging interface through JP1 (UART) and two tactile switches (SW1 and SW2) for boot control. This enables users to enter bootloader mode conveniently and flash using command line(Please refer Efactor SDK)

Switch	Connected To	Function
SW1	IO9	Hold low during reset to enter bootloader mode
SW2	EN	Toggles enable/reset signal for reboot

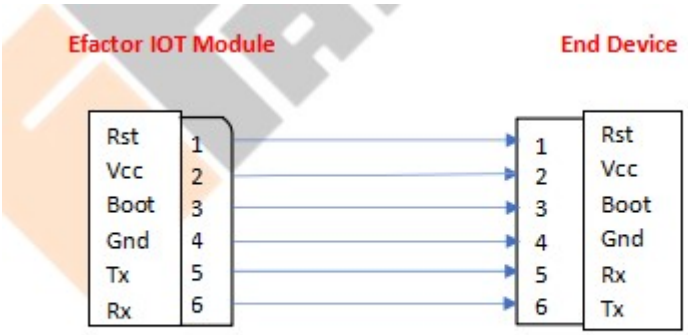
5.1 Flashing Procedure

• Connect a UART adapter to JP1 (TX ↔ RX, RX ↔ TX, GND ↔ GND, VCC↔ VCC).
• Press and hold SW1.
• Press and release SW2 (EN) to reset the module while SW1 is held low.
• Release SW1 — the device now enters bootloader mode.
• After flashing, press SW2 (EN) again to reboot into normal operation.

6. Pin Description

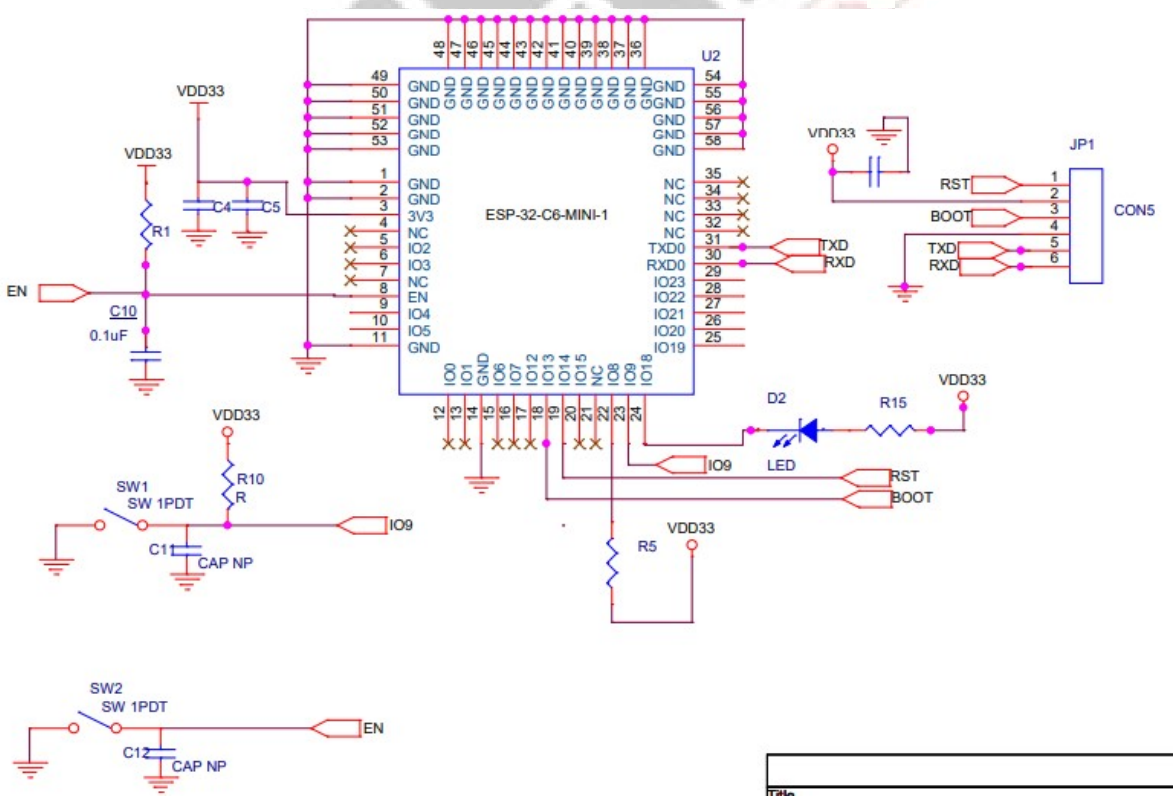
S.No	Pin	Description
1	Reset	Optional reset pin
2	VCC	3.3 V power input (0.5 A Min)
3	Boot	Optional boot mode selector
4	GND	Common ground
5	TX	UART Transmitter
6	RX	UART Receiver

7. Interface Mapping



E-Factor IoT Module	End Device
RST	Optional
VCC	VCC
BOOT	Optional
GND	GND
RX	TX
TX	RX

8. Reference Schematic



R1,R5,R10: 10k

C11,C12: 0.1uF

C4,C5:22uf,0.1uf

9. Functional Description

- Establishes UART link with end device for bidirectional communication.
- Communicates with E-Factor Gateway/Cloud using E-Factor Protocol.
- Supports interaction with both cloud and mobile app.
- Enables OTA firmware updates directly from E-Factor Cloud.

10.Typical Applications

- Smart Home Appliances
- Industrial Automation Systems
- Lighting Controllers
- Smart Fans / HVAC Modules
- Energy Monitoring Devices
- RF or PLC-based Device Clusters